

Sourabh Joshi

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Professional Summary

Ph.D. scholar in Electrical Engineering (Power Electronics) at IIT Roorkee with an M.Tech in Power Electronics and Drives from VNIT Nagpur. Research interests include power electronic converters, wide-bandgap semiconductor devices (SiC/GaN), battery management systems, and AI-driven energy systems. Proficient in MATLAB/Simulink, Python, and machine learning, with research experience in Li-ion battery state estimation, an IEEE conference publication, and prior industrial experience in electrical maintenance and automation.

Education

Indian Institute of Technology, Roorkee <i>Ph.D. in Electrical Engineering (Power Electronics)</i>	July 2026 – Expected 2030
Visvesvaraya National Institute of Technology (VNIT Nagpur) <i>M.Tech in Power Electronics & Electrical Drives</i>	July 2024 – June 2026 CGPA: 7.20/10
Maulana Azad National Institute of Technology (MANIT Bhopal) <i>B.Tech in Electrical Engineering</i>	July 2017 – May 2021 CGPA: 6.91/10

Experience

Indian Institute of Technology Roorkee (IIT Roorkee) <i>Junior Research Fellow (JRF) – ISRO Sponsored Project</i>	Jul 2026 – Present Roorkee, Uttarakhand
- Working on the ISRO, Bengaluru sponsored research project titled "<i>Enhanced Protection Enabled GaN-Based High Voltage Electronic Power Conditioner for Travelling Wave Tube Amplifiers (TWTAs) for Aerospace Applications</i>" - Conducting research on GaN-based high-voltage power electronic converters and electronic power conditioner (EPC) architectures for aerospace applications. - Designing and analysing high-frequency isolated DC–DC converter topologies, protection techniques, control strategies, and high-reliability power conversion systems for space applications. - Contributing to the development of efficient, compact, and fault-tolerant power conditioning systems for Travelling Wave Tube Amplifiers used in satellite communication payloads.	
MSP Steel and Power Limited <i>Graduate Engineering Trainee, Pellet Division – Electrical Department</i>	Jan 2022 – Jan 2023 Raigarh, Chhattisgarh
- Supported operation, preventive maintenance, and troubleshooting of plant electrical systems in a pellet production environment. - Worked with MCCs, transformers, motors, starters, automation systems, and electrical distribution equipment used in continuous industrial operations. - Assisted fault identification, corrective maintenance coordination, and equipment reliability activities for production-critical electrical assets.	

Projects

Deep Learning Based SoC and SoH Estimation for Li-Ion Battery BMS	July 2025 – April 2026
- Developed AI-based battery state estimation models for Li-ion BMS applications, targeting SoC and SoH prediction under varying operating and aging conditions. - Implemented and compared LSTM, GRU, and hybrid CNN–LSTM architectures using voltage, current, temperature, and degradation-related battery data. - Designed an Enhanced Genetic Algorithm for automated hyperparameter optimization to improve model accuracy, convergence stability, and robustness. - Evaluated models using RMSE, MAE, and R^2 metrics and benchmarked performance against conventional battery state estimation approaches.	
Evaluation of Wide-Bandgap Devices for EV Power Electronics	May 2026 – June 2026
- Evaluated Si, SiC, and GaN power semiconductor devices for EV traction inverters, onboard chargers, DC fast chargers, and auxiliary DC–DC converters. - Analyzed conduction losses, switching losses, thermal behavior, packaging constraints, EMI concerns, and reliability challenges of wide-bandgap devices. - Prepared application-oriented comparison frameworks linking device selection with efficiency, power density, thermal management, and system-level performance.	
BLDC Motor Speed Control Using MATLAB/Simulink	Jan 2020 – May 2020
- Built a closed-loop speed control model for a BLDC motor using a PID controller in MATLAB/Simulink. - Tuned controller parameters to improve transient response, reduce steady-state error, and shorten settling time in simulation. - Implemented a three-phase inverter with Hall-sensor-based electronic commutation and closed-loop speed feedback for motor drive operation.	

Publication

A Comparative Analysis of Various Deep Learning Models for State-of-Charge Estimation in Lithium-Ion Batteries

- Presented at IEEE STPEC 2025, NIT Goa, Dec 10–13, 2025.

Evaluation of Wide Bandgap Devices for Electric Vehicle Power Electronics

- Accepted for publication at the 10th IEEE IICPE 2026, MANIT Bhopal, India, Dec 13–15, 2026.

Precise State-of-Health Estimation for Lithium-Ion Batteries Using EGA-Optimized Spatiotemporal Deep Learning

- Accepted for publication at the 10th IEEE IICPE 2026, MANIT Bhopal, India, Dec 13–15, 2026.

Certifications

Supervised Machine Learning: Regression and Classification, DeepLearning.AI
Coursera Certificate May 2025

Introduction to Power Electronics, University of Colorado Boulder
Coursera Certificate Mar 2025

Arena FIDE Master, International Chess Federation (FIDE) Dec 2021

Technical Skills

Programming and Control: Python, MATLAB, C, DSP programming, real-time control implementation

Simulation and Modeling: PLECS, MATLAB/Simulink, circuit-level simulation, converter modeling, control-system modeling, machine learning, deep learning, data preprocessing, model validation

Power Electronics: DC–DC converters, isolated DC–DC converters, high-frequency converters, resonant converters, inverters, PWM techniques, soft switching, high-frequency switching, power converter loss analysis

Magnetics and Converter Design: High-frequency transformer design, planar transformer concepts, inductor design, magnetic component selection, core selection, winding design, flux-density analysis, core and copper loss estimation

Wide-Bandgap Power Devices: GaN HEMTs, SiC MOSFETs, wide-bandgap semiconductor devices, high-frequency switching, gate-drive requirements, switching and conduction loss analysis

Digital Control and Embedded Systems: TI C2000 microcontrollers, TMS320F28379D, DSP programming, ePWM generation, ADC interfacing, dead-time generation, real-time control, digital control implementation

Hardware Design and Prototyping: PCB design, schematic development, PCB layout, component selection, SMD components, through-hole components, soldering, hardware debugging, oscilloscope-based measurements, power-stage prototyping

Battery and Energy Systems: Li-ion batteries, battery management systems, battery energy storage systems, SoC estimation, SoH estimation, EV charging systems

Power Electronics and Drives: Electrical drives, motor control, BLDC motors, inverter-fed drives, control systems, converter-fed motor drives

Industrial Electrical Systems: MCCs, transformers, motors, automation systems, electrical troubleshooting, preventive maintenance

Achievements, Memberships, and Languages

- GATE 2026: All India Rank 4032
- GATE 2024: All India Rank 5151
- JEE Main 2017: All India Rank 14780
- IEEE Graduate Student Member and IEEE Power Electronics Society (PELS) Member
- Languages: English, Hindi, Dutch